

The Closed SIR Flow for All Positive Parameters: Phase Curves, Final-Size Branches, and the No-Recurrence Boundary

Route-A structural certificate C198

Abstract

We give one complete dynamical atlas for the closed mass-action SIR system at all positive transmission and removal rates. Exact scaling reduces the family to one planar flow; its first integral yields the phase curves, infection peak, time quadrature, final-size sensitivity and the correctly selected Lambert branch. A monotone removed coordinate proves global convergence and excludes every nonconstant recurrent orbit. A Lambert-free high-precision checker independently reconstructs both real final-size intersections. The result is an idealized mathematical certificate, with no clinical data or medical advice. Its monotonicity and absence of intrinsic prime arithmetic force strict Route-A rejection.

Keywords: SIR model; phase portrait; final size; Lambert W; threshold; monotone dynamics; Route-A obstruction.

中文摘要

本文对全部正传播率与移除率的封闭质量作用SIR系统给出统一动力学图谱。精确无量纲化把全参数族化为一个平面流；第一积分同时确定相曲线、感染峰值、时间积分、最终规模灵敏度及正确的Lambert W分支。移除变量严格单调，因而全局收敛并排除全部非常值回归轨道。本文只研究理想化数学模型，不使用临床数据，也不提供医疗建议。
关键词：SIR模型；相图；最终规模；阈值；单调动力学。

1 One phase portrait for all parameters

Fix $\beta, \gamma > 0$ and physical time t in

$$\dot{S} = -\beta SI, \quad \dot{I} = \beta SI - \gamma I, \quad \dot{R} = \gamma I.$$

The model and its threshold/final-size analysis descend from Kermack and McKendrick [1]; Lambert-branch analysis is treated explicitly by Pakes [2]. Put

$$\kappa = \gamma/\beta, \quad x = S/\kappa, \quad y = I/\kappa, \quad z = R/\kappa, \quad \tau = \gamma t.$$

Every parameter pair is thereby conjugate, without changing physical-time orientation, to

$$x' = -xy, \quad y' = y(x-1), \quad z' = y. \quad (1)$$

The nonnegative fixed-population simplex is invariant and compact, so solutions are global. Direct differentiation gives

$$H(x, y) = x + y - \log x = H(x_0, y_0), \quad y = y_0 + x_0 - x + \log(x/x_0). \quad (2)$$

For $y_0 > 0$, x decreases strictly. If $x_0 > 1$, then y grows until the unique crossing $x = 1$ and

$$y_{\max} = y_0 + x_0 - 1 - \log x_0.$$

If $x_0 < 1$, y decreases from the start; at $x_0 = 1$ its first derivative is zero, after which decreasing x makes it negative. Thus the threshold and peak are globally, not locally, classified.

2 Global limit, branch, and stability

Since $z' = y \geq 0$ and population is bounded, $\int_0^\infty y d\tau < \infty$; bounded derivatives give $y(\tau) \rightarrow 0$. Also $x(\tau) = x_0 \exp(-\int_0^\tau y ds) > 0$. Hence every $y_0 > 0$ trajectory has $0 < x_\infty < \min(x_0, 1)$ and (2) gives

$$x_\infty e^{-x_\infty} = x_0 e^{-x_0 - y_0}, \quad x_\infty = -W_0(-x_0 e^{-x_0 - y_0}). \quad (3)$$

The function $x - \log x$ decreases on $(0, 1)$ and increases on $(1, \infty)$. Thus $-W_{-1}$ is the other, upper intersection of the invariant curve; it is not the forward limit. The exact time representation is

$$\tau = \int_{x(\tau)}^{x_0} \frac{du}{u\{y_0 + x_0 - u + \log(u/x_0)\}}.$$

Implicit differentiation of (3) yields $\partial x_\infty / \partial y_0 = x_\infty / (x_\infty - 1) < 0$.

If $y_0 = 0$, the initial point is already an equilibrium and remains fixed. In particular, when $x_0 > 1$ its final value is the upper root x_0 , not the positive-infection W_0 branch. On each population simplex the equilibrium line has tangential eigenvalue zero and transverse physical eigenvalue $\beta S_* - \gamma$. Finally, R is strictly increasing wherever $I > 0$; there is no nonconstant periodic or recurrent orbit, and every trajectory converges to the disease-free line.

3 Independent certificate and Route-A stop

The exact release uses 24 positive-infection phase curves: nine subcritical, three threshold and twelve supercritical starts, plus four physical scalings. The producer evaluates 48 real Lambert-branch values. The independent checker calls no Lambert routine: 100-digit Decimal logarithms and monotone bisection recover both intersections. A separate symbolic path proves seven structural identities and tests every branch equation. Replay is byte exact; twelve repaired-hash attacks and one stale-hash attack are rejected. This ledger tests conventions and does not replace the all-parameter proof.

The flow has no nonconstant primitive-periodic-orbit layer, rational-prime carrier, prime-power repetition or $\log p$ clock. Its terminal Lambert equation is not a dynamical Zeta or target Fredholm determinant, and the dissipative flow has no source-native same-clock unitary lift. Therefore

$$(A0, A1, A2, A3, A4) = (\text{FAIL}, \text{FAIL}, \text{FAIL}, \text{FAIL}, \text{FAIL}),$$

overall ROUTE_A_REJECTED, Route B false, under NO_BAD_EULER_OR_ROOT_NUMBER.

Safety and scope firewall. This idealized model supplies no prediction, calibration, intervention advice or claim about a real outbreak. We claim no classical priority, elementary explicit formula for all components versus time, target zero/prime data, arithmetic local datum, Euler factor, root number, automorphy, target analytic structure, Hilbert–Pólya operator, external review or acceptance score.

Revision focus. Round 2 adds the zero-infection boundary, independent branch validation, source ownership, safety and Route-A firewalls.

Declarations

Data and code availability. Package-local synthetic exact evidence and deterministic code accompany this manuscript; no clinical data are used.

Ethics. No human, animal, clinical, personal or sensitive data are used; approval is not applicable.

Author contributions (CRediT). This anonymous certificate records Conceptualization, Formal analysis, Software, Validation, Writing—original draft, and Writing—review and editing. AI systems are not authors.

Funding. No external funding is reported.

Conflicts of interest. None known.

AI-use disclosure. An AI coding assistant supported drafting and exact-code development; it was not an external reviewer or independent error process.

References

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